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ROAD ACCIDENT ANALYSIS USING MICROSOFT POWER BI DESKTOP

- **An Interactive Business Intelligence Project Report**

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1. Abstract

Traffic accidents are among the biggest issues faced by everyone around the globe, as each year millions of humans and their properties are losing their lives due to accidents. It can be easily noticed that more and more vehicles are increasing day by day in our cities and towns. This is why; it becomes highly essential for us to identify and analyze the trend of accidents to make sure that everything goes fine on the road. For this reason, the current research work focuses on making a **Road Accident Analysis Dashboard** using Microsoft Power BI.

This dashboard contains information regarding various parameters that are associated with accidents including total number of accidents, death rate, involvement of different vehicles based on their type and weather conditions etc. Additionally, last year's data has been compared to show how much improvement has been made in the implementation of safety measures.

The main objective behind the analysis is to provide useful insights about the issue and suggest some prevention measures to avoid any future accidents.

2. Introduction

The issue of road accidents around the world is prevalent, leading to a number of losses in terms of human lives as well as destruction of properties. Such incidents happen due to multiple reasons including increased traffic, deteriorated roads, and human errors. In order to get to the bottom of the matter and take corrective actions, it becomes important to analyze the data related to such accidents.

Most data analysis methods prove to be ineffective when it comes to delivering prompt insights on any type of data. However, with Business Intelligence tools like Microsoft Power BI, the data can be analyzed in a more effective manner through the use of dashboards.

This projects discusses the development of a **Road Accident Analysis Dashboard** with an overview of the critical features included in it.

3. Problem Statement

Road accidents are a major public safety concern worldwide, leading to significant loss of life and property. Governments and transport authorities collect vast amounts of accident-related data, including casualty severity, vehicle types, road conditions, and environmental factors.

However, raw datasets alone are difficult to interpret and do not provide actionable insights. The challenge is to transform this complex accident data into a meaningful, interactive visualization that helps identify patterns, high-risk factors, and areas requiring intervention.

This dashboard aims to present a comprehensive overview of road accident trends, enabling stakeholders to quickly understand accident severity, contributing factors, and temporal patterns.

4. Business Use (Real-World Purpose)

This dashboard supports decision-making for:

- **Government Authorities:** To design road safety policies
- **Traffic Management Departments:** To identify high-risk zones and conditions

- **Urban Planners:** To improve infrastructure planning
- **Public Awareness Campaigns:** To target key risk factors (e.g., night driving, rural roads)

It enables quick identification of accident trends and helps prioritize safety measures effectively.

5. Solution & Dashboard Features

The dashboard is a single-page interactive Power BI report, designed for quick insights. Below is a breakdown of each component:

KPI Cards (Top Summary Metrics)

- **Total CY Casualties:** 195.7K (decreased by **11.9%**)
- **Total CY Accidents:** 108.0K (decreased by **11.9%**)
- **Fatal Casualties:** 2.9K (decreased by **33.3%**)
- **Serious Casualties:** 27.0K (decreased by **16.2%**)
- **Slight Casualties:** 165.8K (decreased by **10.6%**)

This indicates an overall decline in accidents and casualties, especially fatal cases.

Casualties by Vehicle Type

Breakdown of casualties based on vehicle involvement:

- **Car:** 155,804 (highest contribution)
- **Bike:** 15,610
- **Van:** 15,905
- **Bus:** 6,573
- **Agricultural Vehicles:** 399
- **Other:** 1,446

Cars dominate accident involvement, indicating higher exposure and usage.

Monthly Trend (CY vs PY Casualties)

- Displays comparison between **Current Year (2022)** and **Previous Year (2021)**
- Shows seasonal fluctuations across months
- General observation:
    - Higher casualties in mid-year (May–October)
    - Drop towards December

Helps identify seasonal accident trends and peak-risk months.

Casualties by Urban vs Rural Areas

- **Urban:** 61.95%

- **Rural:** 38.05%

Urban areas have higher accident rates, likely due to:

- Higher traffic density
- Increased vehicle interactions

Casualties by Road Type

- **Single Carriageway:** 145K (dominant)
- **Dual Carriageway:** 32K
- **Roundabout:** 13K
- **One-way Street:** 3K
- **Slip Road:** 3K

Single carriageways are the most accident-prone, indicating need for safety improvements.

Casualties by Light Conditions

- **Daylight:** ~74%
- **Dark:** ~26%

Although most accidents occur during the day (due to higher traffic),
night conditions still pose significant risk per vehicle density.

Filters / Slicers

- Road Surface Condition
- Weather Condition
- Local Authority (District)

This enables dynamic exploration of accident data by region and environment.

6. Dashboard Screenshot

Figure - Road Accident Analysis Dashboard Overview (Power BI)

7. Tech Stack & Dataset

Area	Details
Visualization Tool	Microsoft Power BI Desktop (2024)
Data Source	CSV File (Road Accident Dataset)
Data Transformation	Power Query Editor
Calculations	DAX Measures (Totals, YoY %, Distribution)
Chart Types	KPI Cards, Line Chart, Bar Chart, Donut Chart
Interactivity	Filters, Slicers, Cross-filtering
File Format	.pbix

8. Key Insights & What Stands Out

a) Overall Improvement in Road Safety

- Total accidents and casualties decreased by approximately 12%
- Fatal casualties reduced significantly (approx. 33%)

Suggests improved safety measures or regulations.

b) Cars Contribute the Most to Casualties

- Cars account for the majority of accidents
  Indicates focus area for:
  - Driver awareness
  - Traffic regulation enforcement

c) Urban Areas are More Accident-Prone

- Approximately 62% accidents occur in urban regions due to congestion, intersections, and dense traffic

d) Single Carriageways are High-Risk

- Largest share of accidents (145K), suggests:
  - Need for better road design
  - Lane separation or traffic control

e) Daytime Accidents Dominate

- Majority accidents occur during daylight due to higher traffic volume, not necessarily higher risk per vehicle

f) Seasonal Trends Exist

- Mid-year months show higher accident frequency
  Useful for:
  - Planning awareness campaigns
  - Deploying traffic control measures

9. Future Scope & Conclusion

Future Enhancements

i) Add geospatial maps to visualize accident hotspots.
ii) Include time-of-day analysis (hourly breakdown).
iii) Integrate driver behavior data (speed, alcohol influence).
iv) Add predictive modeling (ML) for accident risk forecasting.
v) Drill-through reports for district-level deep analysis.

Conclusion

This dashboard successfully transforms raw accident data into an intuitive and actionable visual format. Key patterns such as:

- Dominance of car-related accidents
- Urban concentration of casualties
- High risk on single carriageways
- Significant reduction in fatalities

become immediately visible.

The true value of this dashboard lies in its ability to make critical safety insights clear, accessible, and actionable, helping stakeholders make informed decisions to reduce road accidents and save lives.